

OUTPUT FOR TASK 4:

mean radius mean texture mean perimeter mean area mean smoothness \

0	17.99	10.38	122.80	1001.0	0.11840
1	20.57	17.77	132.90	1326.0	0.08474
2	19.69	21.25	130.00	1203.0	0.10960
3	11.42	20.38	77.58	386.1	0.14250
4	20.29	14.34	135.10	1297.0	0.10030

mean compactness mean concavity mean concave points mean symmetry \

0	0.27760	0.3001	0.14710	0.2419
1	0.07864	0.0869	0.07017	0.1812
2	0.15990	0.1974	0.12790	0.2069
3	0.28390	0.2414	0.10520	0.2597
4	0.13280	0.1980	0.10430	0.1809

mean fractal dimension ... worst radius worst texture worst perimeter \

0	0.07871	...	25.38	17.33	184.60
1	0.05667	...	24.99	23.41	158.80
2	0.05999	...	23.57	25.53	152.50
3	0.09744	...	14.91	26.50	98.87
4	0.05883	...	22.54	16.67	152.20

worst area worst smoothness worst compactness worst concavity \

0	2019.0	0.1622	0.6656	0.7119
1	1956.0	0.1238	0.1866	0.2416
2	1709.0	0.1444	0.4245	0.4504
3	567.7	0.2098	0.8663	0.6869
4	1575.0	0.1374	0.2050	0.4000

worst concave points worst symmetry worst fractal dimension

0	0.2654	0.4601	0.11890
1	0.1860	0.2750	0.08902
2	0.2430	0.3613	0.08758
3	0.2575	0.6638	0.17300
4	0.1625	0.2364	0.07678

[5 rows x 30 columns]

0 0
1 0
2 0
3 0
4 0

dtype: int64

Shape of Features: (569, 30)

Shape of Target: (569,)

Target Classes:

1 357

0 212

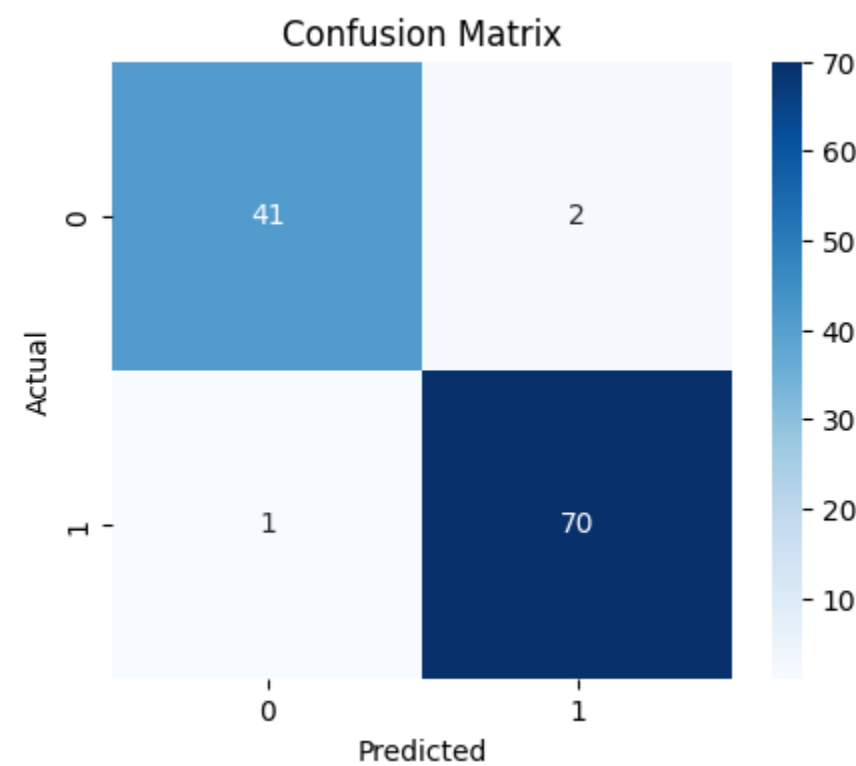
Name: count, dtype: int64

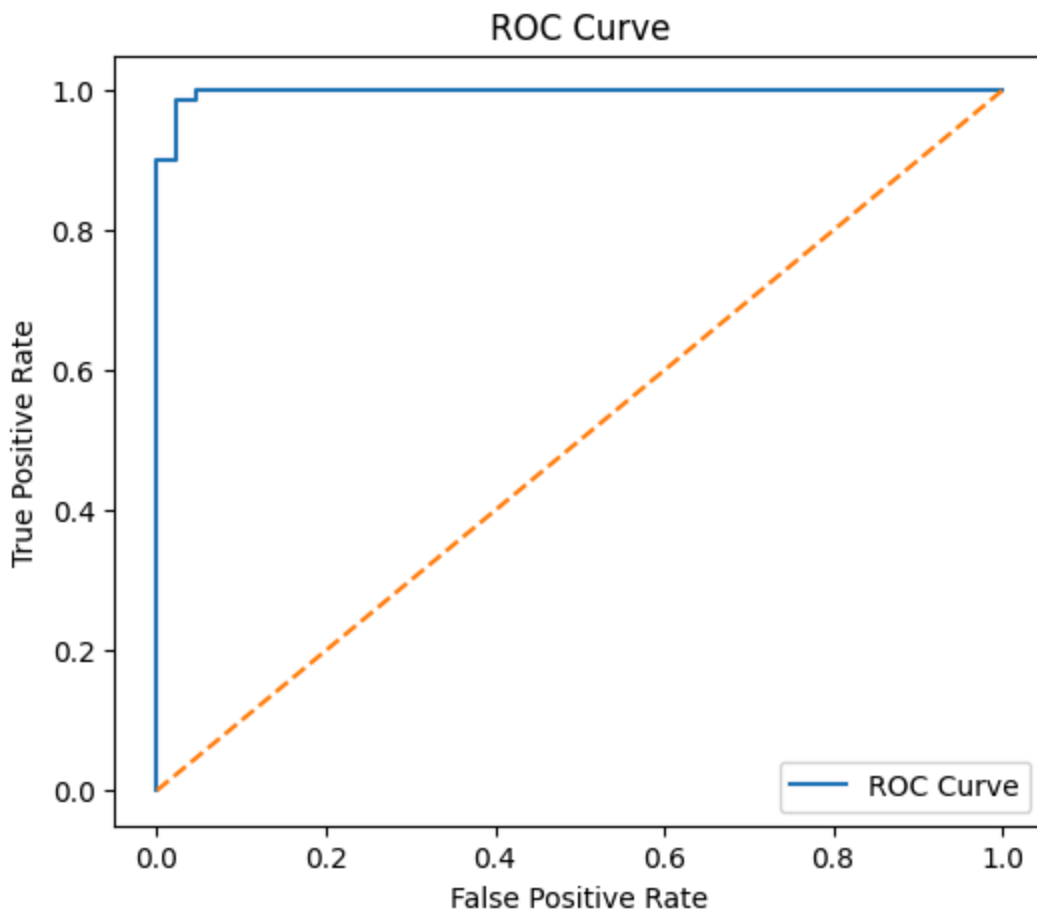
Accuracy: 0.9736842105263158

Precision: 0.9722222222222222

Recall: 0.9859154929577465

	precision	recall	f1-score	support
0	0.98	0.95	0.96	43
1	0.97	0.99	0.98	71
accuracy			0.97	114
macro avg	0.97	0.97	0.97	114
weighted avg	0.97	0.97	0.97	114





ROC-AUC Score: 0.99737962659679

Threshold: 0.4

[[41 2]

[0 71]]

	precision	recall	f1-score	support
0	1.00	0.95	0.98	43
1	0.97	1.00	0.99	71
accuracy			0.98	114
macro avg	0.99	0.98	0.98	114
weighted avg	0.98	0.98	0.98	114

